Data communications (Lab 4): Amplitude Modulation (AM)

Objective: To generate and analyze an Amplitude Modulated (AM) signal using MATLAB by modulating a message signal with a high-frequency carrier signal.

Theory

Amplitude Modulation (AM) is an analog modulation technique in which the amplitude of the carrier signal varies according to the message (modulating) signal, while the frequency remains constant.

In AM:

- Message signal → low frequency
- Carrier signal → high frequency
- Output → amplitude varies according to message

Mathematically:

$$y(t) = m(t).c(t)$$

Where:

- $m(t)$ = message signal
- $c(t)$= carrier signal

The carrier's amplitude follows the shape of the message signal.

```
clear;
clc;

t = 0:0.0005:1;     % Time vector

% Equation of modulating (message) signal
ym = 5*sin(2*pi*5*t);
subplot(3,1,1);
plot(t,ym), grid on;
title('Modulating Signal');
xlabel('Time (sec)');
ylabel('Amplitude');

% Equation of carrier signal
yc = 5*sin(2*pi*100*t);
subplot(3,1,2);
plot(t,yc), grid on;
title('Carrier Signal');
xlabel('Time (sec)');
ylabel('Amplitude');
```

```
% Amplitude modulated signal
y = ym .* yc;
subplot(3,1,3);
plot(t,y);
title('Amplitude Modulated Signal');
xlabel('Time (sec)');
ylabel('Amplitude');
grid on
```

Code Explanation

1. Initialization

```
clc;
clear;
close all;
```

- `clc` → Clears command window
- `clear` → Removes all variables

2. Time Vector

```
t = 0:0.0005:1;
```

- Defines time from 0 to 1 second
- Small step size → smooth waveform

3. Message Signal

```
ym = 5*sin(2*pi*5*t);
```

- Low frequency signal (5 Hz)
- Represents the information signal

4. Carrier Signal

```
yc = 5*sin(2*pi*100*t);
```

- High frequency signal (100 Hz)
- Used to carry the message

5. AM Modulation

```
y = ym .* yc;
```

- Multiplies message and carrier
- Produces amplitude modulated signal

6. Plotting

Subplot 1 → `message signal`
Subplot 2 → `carrier signal`
Subplot 3 → `AM signal`

Result:

The MATLAB program produces:
Message signal (low frequency)
Carrier signal (high frequency)

Observations:

The amplitude of the carrier varies according to the message signal
The shape of the message signal appears as an envelope in the AM waveform

Conclusion: The experiment successfully demonstrated Amplitude Modulation using MATLAB. The message signal was used to vary the amplitude of the carrier signal, and the output waveform clearly showed the envelope of the message signal, verifying the principle of AM.